👩🏾‍🏫 Staff Guide · Horticulture 1 · Unit 3 Parts of the Plant (Level 3)

Level 3 = resource-room students who can work close to the gen ed lesson. No slide decks: students follow the class teacher's lesson and slides with Level 3 readings, guided notes and a quick check.

1. How the pieces line up

Gen ed lesson	Level 3 piece
Her readings and worksheets	Level 3 reading (5 short paragraphs) + check questions
Class notes	Level 3 guided notes + key
Unit quiz / test	Level 3 quick check + key
Review (Kahoot / Jeopardy)	1 Level 3 game on the class site: 10 questions + a matching round

2. Supports built in

- 🔊 Read aloud on the reading and notes pages (tap the button)
- Key terms listed at the top of the notes
- Short-answer lines sized for 1-2 sentences

3. What to grade

- Notes and reading checks: same points as her worksheets for that lesson
- Quick check: score, then scale to her test points
- Modified grading only as the IEP allows. Check with the case manager.

4. Who to ask

- Mr. Driver (SLE): changes and IEP questions.
- Facts we could not confirm say “check with the class teacher”.

🚧 Under construction (check with the class teacher)

- **Pollination mini videos**: This page is empty in the export.
- **Silence of the Bees**: Full PBS episode upload on YouTube; staff should use a short official PBS clip.

💗 Note

Bee safety: check health plans for bee-sting allergies before outdoor or greenhouse work near flowers.

⚠️ Safety: tools and plants

- **Tools**: students may use all tools (pruners, knives, floral wire, low-temp glue guns) **with an adult right there**. Staff hand tools out, stay at the station, and count them back in. Cut-resistant gloves if possible.
- **Poinsettia sap** is milky and can irritate skin and eyes. Gloves on, no rubbing eyes, wash hands after.
- **Latex**: poinsettias are in the latex family. Students with a latex allergy wear non-latex (nitrile) gloves or skip handling; check health plans.
- **Bees and pollen**: check IEP/health plans for bee-sting or pollen allergies and know where the EpiPen is before greenhouse work.
- **Mouths**: no plants, soil, seeds, berries, fertilizer or rooting hormone in mouths. Students who mouth objects or have pica need one-to-one eyes in the greenhouse. Many plants (holly and mistletoe berries, some houseplants) are poisonous.
- **Chemicals**: rooting hormone and fertilizer with gloves, then wash hands. Pesticides and growth regulators are staff/teacher only; students never spray or handle them.
- **Greenhouse**: wet floors and hoses (slips), heat (water breaks), wet dry potting mix first (dust).

🌿 Parts of the Plant: Guided Notes

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

Use the class slides and your reading. Key words:

taproot | *fibrous root* | *root hair* | *xylem* | *phloem* | *petiole* | *stomata* | *transpiration* | *photosynthesis* | *stamen* | *pistil* | *pollination*

A. Roots

1. Root _____ absorb water.
2. Compare a taproot and fibrous roots with an example of each.

B. Stems and leaves

3. _____ moves water up; _____ moves food.
4. Gas exchange happens through the _____.
5. Fill in the table.

Leaf description	Examples
Form	
Margin	
Arrangement	

C. Photosynthesis and flowers

6. Write photosynthesis in words (inputs → outputs).
7. Name the parts of the stamen and the pistil.

🌿 Parts of the Plant: Reading

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

How a Plant Works

Roots anchor a plant in the soil and absorb water and minerals through tiny root hairs. A taproot is one thick main root, like a carrot; fibrous roots are many thin roots, like grass. Some roots store food, and some can be used for propagation. The root cap protects the growing tip as it pushes through soil; it does not absorb water.

Stems hold up leaves and flowers and move materials. Xylem carries water and minerals upward from the roots. Phloem carries food and minerals throughout the plant. In dicot stems, a cambium layer separates the xylem from the phloem in rings; in monocots the vascular bundles are scattered.

Leaves are attached by a petiole, and their main job is photosynthesis. Using light, leaves combine water and carbon dioxide to make sugar (food) and give off oxygen. Gas exchange happens through stomata, tiny openings controlled by guard cells. Through transpiration, leaves give off water vapor. Leaves are described by form (needle, round, ovate), margin (dentate, undulate, incised) and arrangement (opposite, alternate, whorled).

Flowers are the reproductive part. The stamen (male) is made of the anther and filament and produces pollen. The pistil (female) is made of the stigma, style, ovary and ovules. Pollination, often by bees, results in seeds and fruit. Because so many crops depend on bees, declining bee populations are a serious problem for growers.

Check your reading

1. Compare the jobs of xylem and phloem.

2. What are the inputs and products of photosynthesis?

3. Why are bees important to horticulture?

🌸 Flower Dissection: Lab Sheet

Horticulture 1 · Unit 3 · Level 3 · goes with Flowers Worksheet / flower dissection

Name: _____

Question: What are the reproductive parts of a flower and what does each do?

Materials

flower (lily/alstroemeria), knife (staff-supervised), hand lens, parts mat

⚠️ Safety

- Knife only with an adult at the table; cut away from you
- Gloves; wash hands
- Tell staff about pollen allergies

Prediction

How many stamens do you think your flower has?

_____

_____

Procedure

1. Remove sepals and petals.
2. Remove and count stamens.
3. Identify stigma, style and ovary.
4. With staff, cut the ovary and find ovules.
5. Label all parts.

Data

Part	Male/female/other	Job
Petal		
Stamen		
Pistil		
Ovule		

Analysis (Claim, Evidence, Reasoning)

1. Claim: which flower part becomes the seed and which becomes the fruit? Use your observations.

🌿 Leaf Collection Walk: Lab Sheet

Horticulture 1 · Unit 3 · Level 3 · goes with Leaf Collection

Name: _____

Question: How can leaves be classified by form, margin and arrangement?

Materials

8 approved leaves, hand lens, leaf chart, tape

⚠️ Safety

- Gloves on
- Collect only staff-approved plants; no berries
- Wash hands after

Prediction

Which leaf form do you think will be most common in our greenhouse?

_____

_____

Procedure

1. Collect 8 approved leaves.
2. Classify each by form, margin and arrangement.
3. Record in the table.
4. Count which form is most common.

Data

Leaf #	Form	Margin	Arrangement
1			
2			
3			
4			

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was the most common leaf form? Evidence from your table, and why classifying leaves helps identify plants.

🌿 Parts of the Plant: Quick Check

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The root cap's main job is…

A. Water absorption | B. Photosynthesis | C. Protecting the tip

2. Phloem is responsible for…

A. Moving food and minerals | B. Moving only water up | C. Making pollen

3. Undulate, incised and dentate describe…

A. Leaf forms | B. Leaf arrangements | C. Leaf margins

4. The products of photosynthesis are…

A. Oxygen and food | B. Water and CO2 | C. Food and water

5. What is transpiration?

6. What two things result from pollination?

🌿 Parts of the Plant: Guided Notes · ANSWER KEY

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

Use the class slides and your reading. Key words:

taproot · *fibrous root* · *root hair* · *xylem* · *phloem* · *petiole* · *stomata* · *transpiration* · *photosynthesis* · *stamen* · *pistil* · *pollination*

A. Roots

1. Root **hairs** absorb water.

2. Compare a taproot and fibrous roots with an example of each.
Taproot: one thick root (carrot). Fibrous: many thin roots (grass).

B. Stems and leaves

3. **Xylem** moves water up; **phloem** moves food.

4. Gas exchange happens through the **stomata**.

5. Fill in the table.

Leaf description	Examples
Form	**needle, round, ovate**
Margin	**dentate, undulate, incised**
Arrangement	**opposite, alternate, whorled**

C. Photosynthesis and flowers

6. Write photosynthesis in words (inputs → outputs).
Water + carbon dioxide + light → sugar + oxygen.

7. Name the parts of the stamen and the pistil.
Stamen: anther, filament. Pistil: stigma, style, ovary, ovules.

🔑 *Key written by our team from the class teacher's worksheets and slides, please check.*

🌸 Flower Dissection: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 3 · Level 3 · goes with Flowers Worksheet / flower dissection

Name: _____

Question: What are the reproductive parts of a flower and what does each do?

Materials

flower (lily/alstroemeria), knife (staff-supervised), hand lens, parts mat

⚠️ Safety

- Knife only with an adult at the table; cut away from you
- Gloves; wash hands
- Tell staff about pollen allergies

Prediction

How many stamens do you think your flower has?
Lilies and alstroemeria have 6.

Procedure

1. Remove sepals and petals.
2. Remove and count stamens.
3. Identify stigma, style and ovary.
4. With staff, cut the ovary and find ovules.
5. Label all parts.

Data

Part	Male/female/other	Job
Petal	**other**	**attract pollinators**
Stamen	**male**	**makes pollen**
Pistil	**female**	**receives pollen**
Ovule	**female**	**becomes seed**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: which flower part becomes the seed and which becomes the fruit? Use your observations.
Ovules become seeds; the ovary becomes the fruit.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🌿 Leaf Collection Walk: Lab Sheet · ANSWER KEY

Horticulture 1 · Unit 3 · Level 3 · goes with Leaf Collection

Name: _____

Question: How can leaves be classified by form, margin and arrangement?

Materials

8 approved leaves, hand lens, leaf chart, tape

⚠️ Safety

- Gloves on
- Collect only staff-approved plants; no berries
- Wash hands after

Prediction

Which leaf form do you think will be most common in our greenhouse?
Answers vary; ovate is common.

Procedure

1. Collect 8 approved leaves.
2. Classify each by form, margin and arrangement.
3. Record in the table.
4. Count which form is most common.

Data

Leaf #	Form	Margin	Arrangement
1	**ovate**	**dentate**	**alternate**
2	**needle**	**smooth**	**whorled**
3	**round**	**undulate**	**opposite**
4	**ovate**	**smooth**	**alternate**

Expected results (your class data may differ).

Analysis (Claim, Evidence, Reasoning)

1. Claim: what was the most common leaf form? Evidence from your table, and why classifying leaves helps identify plants.
Answers vary; classification narrows down plant ID.

🔑 *Key written by our team from the class teacher's lab sheets, please check.*

🌿 Parts of the Plant: Quick Check · ANSWER KEY

Horticulture 1 · Unit 3 · Level 3 (closer to the class work) · goes with Leaves/Stems/Roots/Flowers worksheets, leaf collection, photosynthesis, Unit 3 quiz

Name: _____

Multiple choice and short answer. You may have it read aloud.

1. The root cap's main job is…

A. Water absorption | B. Photosynthesis | **C. Protecting the tip**

2. Phloem is responsible for…

A. Moving food and minerals | B. Moving only water up | C. Making pollen

3. Undulate, incised and dentate describe…

A. Leaf forms | B. Leaf arrangements | **C. Leaf margins**

4. The products of photosynthesis are…

A. Oxygen and food | B. Water and CO2 | C. Food and water

5. What is transpiration?

The leaf giving off water vapor through stomata during gas exchange.

6. What two things result from pollination?

Seed and fruit.

🔑 *Reworded from her Unit 3 Parts of the Plant quiz (28 Qs with key), please check.*